

$N = 2$ -preserving selection rules for the flow $k \rightarrow k - 2$ and the massive vacua of the massless phase

Working note

Setup.

- We use the bosonic $N = 2$ coset with fields (l, m, s) , the charge-conjugation completion C , and the S-matrix of N2Gab eq. (2.6), normalised by $1/K$ with $K = k + 2$. Throughout, k is odd.
- For a node $T \subset C$ (a closed set of fields, selection.pdf §2.3), F_T is the set of Verlinde lines commuting with T . Its global index is $\mu_T = (\sum_{F_T} d^2)^2$ (paul.pdf (5)), and $\lambda_T = \sqrt{\mu_T}$ is the index of $T \subset C$.
- All numbers come from `n2.susy.py` and `n2.nodes.py`, run for $k = 3, \dots, 11$.

1 The extra rule: $N = 2$ -closed nodes

In the bosonic model the supercurrents act on local fields only through the diagonal field

$$(0, 0, 2) \ni G^\pm \bar{G}^\pm, G^\pm \bar{G}^\mp \quad (h = \bar{h} = 3/2), \quad (1)$$

because $(0, 0, 2) \times (l, m, s) = (l, m, s + 2)$ maps a field to its superpartner sector.

We call a node $N = 2$ -closed if, with every (l, m, s) , it contains $(l, m, s + 2)$. Equivalently, it contains $(0, 0, 2)$. The neutral algebra of an $N = 2$ -preserving flow must be $N = 2$ -closed.

The perturbation is the F-term $G_{-1/2}^- \bar{G}_{-1/2}^-$ of the chiral primary $(k, k, 0) \sim (0, -2, 2)$, and it sits in $(k, k, 2) \sim (0, -2, 0)$ (plus its conjugate). Adding its superconformal primary already forces closure, since $(0, -2, 2) \times (0, 2, 0) = (0, 0, 2)$.

2 UV: the smallest $N = 2$ -closed node of $(0, -2, 0)$

$$N_{UV}^S \equiv \langle (0, -2, 0), (0, -2, 2) \rangle = \{(0, 2n, 0), (0, 2n, 2)\}_{n \in \mathbb{Z}_K}, \quad (2)$$

$$F = \{(L, 0, 0), (L, 0, 2)\}_{L \text{ even}} \simeq \text{so}(3)_k \times \mathbb{Z}_2, \quad (3)$$

$$\mu(N_{UV}^S) = 4 (\dim \text{so}(3)_k)^2, \quad \dim \text{so}(3)_k = \sum_{L \text{ even}} d_L^2 = \frac{K}{4 \sin^2(\pi/K)}. \quad (4)$$

Here $\mathbb{Z}_2 = \{1, L_{(0,0,2)}\}$. The line $L_{(0,0,2)}$ acts as $(-1)^s$, i.e. it is $+1$ on NS fields and -1 on Ramond fields.

Content. N_{UV}^S consists of K NS supermultiplets. Their superconformal primaries $[l, m]$, with $|m| \leq l$, are

$$[0, 0] \quad \text{and} \quad [k, m], \quad m = -k, -k+2, \dots, k, \quad h = \frac{k(k+2) - m^2}{4K}, \quad q = \frac{m}{K}. \quad (5)$$

Each $[l, m]$ contains:

- the bottom component, of dimension $\Delta = 2h$, in $(l, m, 0)$;
- $G_{-1/2}^\pm \bar{G}_{-1/2}^\pm$ components, with $\Delta = 2h + 1$, in $(l, m, 2)$;
- the D-term, with $\Delta = 2h + 2$, which is always irrelevant.

k	relevant, $N = 2$ -preserving	relevant, $N = 2$ -breaking (bottom components)
3	F-terms of $[3, \pm 3]$, $\Delta = 8/5$	$[3, \pm 3]$ ($\Delta = 3/5$), $[3, \pm 1]$ ($\Delta = 7/5$)
5	F-terms of $[5, \pm 5]$, $\Delta = 12/7$	$[5, \pm 5]$ ($5/7$), $[5, \pm 3]$ ($13/7$)
≥ 7	F-terms of $[k, \pm k]$, $\Delta = 2 - 2/K$	$[k, \pm k]$ ($\Delta = k/K$) only

So the only $N = 2$ -preserving relevant perturbation in N_{UV}^S is ϕ_{UV} itself, and it is exclusive to this node.

All $N = 2$ -closed nodes containing ϕ_{UV} . For odd k , every nontrivial line of $\mathfrak{so}(3)_k$ generates all of $\mathfrak{so}(3)_k$ under fusion (checked). The nodes containing N_{UV}^S therefore correspond to the four subrings of F :

node	fields	F	μ
N_{UV}^S	$(0, 2n, s)$, s even	$\mathfrak{so}(3)_k \times \mathbb{Z}_2$	$4 \dim^2 \mathfrak{so}(3)_k$
simple currents	all $(0, m, s)$ (incl. Ramond)	$\mathfrak{so}(3)_k$	$\dim^2 \mathfrak{so}(3)_k$
$\mathcal{NS}_k$	all NS fields (s even)	$\mathbb{Z}_2$	4
C	everything	1	1

The NS fields of the simple-current node are those of N_{UV}^S , so it has no new $N = 2$ perturbation. $\mathcal{NS}_k$ contains the F-terms of all chiral primaries $X^j \leftrightarrow [j, \pm j]$, $j = 1, \dots, k$. Those with $j < k$ are exclusive to it.

3 IR: the $N = 2$ -closed node of the IR operator

The $k - 2$ model is approached by the D-term of the non-BPS $(4, \pm 2, 0)$ (N2Gab §5.3). A D-term is $N = 2$ -preserving and lies in $(4, \pm 2, 0)$ itself. Its $N = 2$ -closed node is

$$\langle (4, 2, 0), (0, 0, 2) \rangle_{k-2} = \mathcal{NS}_{k-2}, \quad F = \mathbb{Z}_2, \quad \mu = 4. \quad (6)$$

The only $N = 2$ -closed IR nodes containing it are $\mathcal{NS}_{k-2}$ ($\mu = 4$) and C_{k-2} ($\mu = 1$). For $k - 2 < 4$, $(4, 2, 0)$ does not exist, but any IR NS operator with $l \neq 0$ again generates $\mathcal{NS}_{k-2}$.

4 Selection rules (selection.pdf §3) with the $N = 2$ constraint

k	$\mu(N_{UV}^S)$	$\mu(\mathcal{NS}_k)$	$\mu(\mathcal{NS}_{k-2})$
3	52.36	4	4
5	345.65	4	4
7	1479.85	4	4
9	4801.50	4	4
11	12880.78	4	4

(a) Perturbation by ϕ_{UV} alone: no massless IR. The flow stays in N_{UV}^S . By transportability, all of $F = \mathfrak{so}(3)_k \times \mathbb{Z}_2$ is preserved at every scale. No IR node matches $\mu = 4 \dim^2 \mathfrak{so}(3)_k$. More strongly, no IR of the form “ $(k-2)$ model $\oplus$ gapped vacua” can carry $\mathfrak{so}(3)_k$:

1. A line cannot connect the $c_{k-2} > 0$ vacuum to a $c = 0$ vacuum, so it acts on the CFT vacuum by a number $\chi(L) = \sum_j n_j d_j^{IR} > 0$.
2. A positive number for every line compatible with fusion is necessarily $\chi = d_L$.
3. Then $d_L \in \mathbb{Q}(\zeta_K) \cap \mathbb{Q}(\zeta_{8(K-2)}) \cap \mathbb{R} = \mathbb{Q}$, since $\gcd(K, K-2) = 1$ for K odd. But $d_2 = 1 + 2 \cos(2\pi/K) \notin \mathbb{Q}$ for odd $K \geq 5$.

The flow is therefore fully massive. It is the Chebyshev flow, which is the unique superpotential keeping $\mathfrak{so}(3)_k$ (N2Gab §4).

(b) The massless flow. The couplings of X^k and X^{k-2} have dimensions $2/K$ and $4/K$, so they are resonant. At $O(\lambda^2)$ the linear LG deformation $X^{k+2} + \lambda X^k$ therefore also switches on the F-term of X^{k-2} , which lies in $(k-2, k-2, 2) \sim (2, -4, 0)$. This field is exclusive to $\mathcal{NS}_k$. This is the CFT form of the second-order obstruction of N2Gab §3.4/§5.2. The neutral algebra becomes

$$\mathcal{NS}_k \longrightarrow \mathcal{NS}_{k-2}, \quad \mu = 4 = 4, \quad (7)$$

and the rule is satisfied. What survives is only the $\mathbb{Z}_2$ line $L_{(0,0,2)}$, which separates the NS from the Ramond sector. $\mathfrak{so}(3)_k$ is not a symmetry of the massless flow.

An extra symmetry that is not in F . The lines $L_{(0,K,\pm 1)}$, with $L_{(0,K,1)}^2 = L_{(0,0,2)}$, act on (l, m, s) by $(-1)^{m_i - s}$. Consequently:

- They act trivially on both perturbations $(0, -2, 0)$ and $(2, -4, 0)$, and on the IR $(4, \pm 2, 0)$. They are therefore exact symmetries of the massless flow.
- They act by -1 on $(0, 0, 2)$ and on the chiral primaries $(j, j, 0)$ with j odd, e.g. $X = (1, 1, 0)$. So they commute with the flow but not with the $N = 2$ -closed node. They are the R -parity $X \rightarrow -X$, $W \rightarrow -W$ (compensated by $\theta \rightarrow i\theta$).
- The largest (not $N = 2$ -closed) node on which they act trivially is $E_k = \{(l, m, 0) : l, m \text{ even}\}$. It has $\mu(E_k) = 16$, and $E_k \subset \mathcal{NS}_k$ with relative index 2.

5 Entropic order parameters and the massive vacua

Language (entropic.pdf §3, modular.pdf §3.7). Take a two-interval region $R = R_1 \cup R_2$ with cross-ratio x . Its complement R' is again two intervals, with cross-ratio $1 - x$. For a pair $T \subset M$ of nodes, define:

- the order parameter $S_{T|M}(x) \equiv S_{M(R)}(\omega|\omega \circ \varepsilon) = I_M(x) - I_T(x)$ (modular.pdf (3.68)), which tests the intertwiners;
- its dual $\tilde{S}_{T|M}(1 - x)$ on R' , which tests the twists.

The certainty relation (entropic.pdf (3.25)) reads

$$S_{T|M}(x) + \tilde{S}_{T|M}(1 - x) = \log \lambda_{T|M}, \quad 0 \leq S, \tilde{S} \leq \log \lambda_{T|M}. \quad (8)$$

In $d = 2$ intertwiners and twists are not distinguished by topology, as they are in $d > 2$ (balls versus shells). Both are non-local operators of two-interval regions, labelled by the same set $F \simeq \text{DHR}$ (paul.pdf), and exchanging them is $R \leftrightarrow R'$, $x \leftrightarrow 1 - x$. What distinguishes phases is the behaviour at large scale for fixed x (entropic.pdf §3.4, §4):

- *massless*: $S(x)$ is a nontrivial conformal function rising from 0 to $\log \lambda$;
- *gapped, sectors unbroken*: $S \rightarrow 0$ (area law) and $\tilde{S} \rightarrow \log \lambda$ (constant law), for all x ;
- *gapped, broken*: $S \rightarrow \log \lambda_{\text{vac}}$ (constant law) and $\tilde{S} \rightarrow \log(\lambda/\lambda_{\text{vac}})$, independent of x .

λ_{vac} is the total quantum dimension of the topological (vacuum-degeneracy) operators, i.e. D_{TQFT}^2 . For invertible lines it is the number of vacua in one orbit. Positivity gives

$$1 \leq D_{\text{TQFT}}^2 = \lambda_{\text{vac}} \leq \lambda_{T|M}. \quad (9)$$

Because λ is scale independent (transportability), a vacuum in which all sectors stay massless requires $\mu_{UV} = \mu_{IR}$. If instead some sectors become massive in the same vacuum, they carry $D_{\text{TQFT}}^2 = \sqrt{\mu_{UV}/\mu_{IR}}$.

Massless flow, CFT vacuum. For $\mathcal{NS}_k \subset C$ we have $\mu_{UV} = \mu_{IR} = 4$, so $\sqrt{4/4} = 1$. No topological operators appear on top of the $(k - 2)$ model in its own vacuum. The massless algebra A_m of massless.pdf contains all of $\mathcal{NS}_{k-2}$.

Massless flow, massive vacua. The TQFT.

- *Pair $\mathcal{NS}_k \subset C$* ($\lambda = 2$; charged fields are the Ramond fields). Nothing in this pair forces extra vacua. In the massive vacua the fermions are massive and the Ramond intertwiners decay (area law), so $S \rightarrow 0$ and $\tilde{S} \rightarrow \log 2$. The line $L_{(0,0,2)}$ is unbroken.
- *Pair $E_k \subset \mathcal{NS}_k$* ($\lambda = 2$; charged fields are the R -odd NS fields, e.g. X). The value of the superpotential at a massive vacuum is $W(v) \neq 0$, and R -parity maps $W \rightarrow -W$. So massive vacua come in pairs exchanged by R -parity, $\langle X \rangle = \pm i\sqrt{\lambda}$. The order parameter $S_{E|\mathcal{NS}}(x) \rightarrow \log 2$ with a constant law, and its dual $\rightarrow 0$.
- *Bounds.* $2 \leq D_{\text{TQFT}}^2 \leq \lambda(E_k \subset C) = 4$. The Witten index $k + 1 = (k - 1) + 2$ fixes $D_{\text{TQFT}}^2 = 2$.

The TQFT consists of the two vacua $X = \pm i\sqrt{\lambda}$. Its topological local operators are $\{1, \text{sign}(X)\}$, with R -parity broken and $L_{(0,0,2)}$ unbroken.

Chebyshev flow (check). In N_{UV}^S , $\lambda = 2 \dim \text{so}(3)_k$ and everything is massive.

- Acting with a line on a vacuum of a gapped phase gives a superposition of vacua with non-negative integer multiplicities, and the largest eigenvalue of that integer matrix is d_L . Since $d_L \notin \mathbb{Z}$, $\text{so}(3)_k$ cannot act on a single vacuum: at least $\deg \mathbb{Q}(d_2) = \varphi(K)/2$ vacua are needed. The regular action needs $(k + 1)/2$ vacua.
- $L_{(0,0,2)}$ is unbroken, and R -parity doubles the vacua.

Together this gives $\frac{k+1}{2} \times 2 = k + 1$ massive vacua, which is exactly the Chebyshev vacuum count (N2Gab §4). The same bound shows that two massive vacua can never host $\text{so}(3)_k$ for $k \geq 5$.

Relation to massless.pdf. The conjecture (massless and massive together $\Rightarrow$ a symmetry charging the massive operators) is realised here across vacua. The symmetry is R -parity $L_{(0,K,\pm 1)}$. It is invisible in the $N = 2$ -closed node (it does not commute with $(0,0,2)$), but it commutes with the flow. The massive vacua are its orbit, and $\text{sign}(X)$ is the massive, charged topological operator.

Summary

1. Smallest $N = 2$ -closed node of $(0, -2, 0)$: $N_{UV}^S = \{(0, 2n, 0), (0, 2n, 2)\}$, i.e. K supermultiplets $[0, 0], [k, m]$ (m odd). Its only $N = 2$ -preserving relevant field is ϕ_{UV} . $F = \text{so}(3)_k \times \mathbb{Z}_2$ and $\mu = 4 \dim^2 \text{so}(3)_k$. It has no massless IR, so perturbing by ϕ_{UV} alone gives the massive Chebyshev flow with $k + 1$ vacua.
2. The massless flow needs the resonant F-term of X^{k-2} , which is exclusive to $\mathcal{NS}_k$. The rule is then $\mathcal{NS}_k \rightarrow \mathcal{NS}_{k-2} = \langle (4, 2, 0), (0, 0, 2) \rangle$, with $\mu = 4$ at both ends. Only $L_{(0,0,2)}$ (NS vs Ramond) survives in F .
3. The TQFT on top of the $(k - 2)$ model is two vacua exchanged by R -parity, $D^2 = 2$, within the entropic bound $2 \leq D^2 \leq 4$. There is no topological sector inside the CFT vacuum.

Caveats.

- Odd k only. For even k the line $[k, 0, 0]$ is invertible and survives (N2Gab App. D).
- “Flow inside $N_{UV}^S = \text{Chebyshev}$ ” uses the uniqueness statement of N2Gab §4.
- The field argument assumes that IR lines have dimensions in the cyclotomic field of the $(k - 2)$ model.
- The Witten-index count of the massive vacua is LG input, not a consequence of the node analysis.